JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY KAKINADA

DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING

ELECTRICAL MACHINES & MEASUREMENTS LAB

★ LABORATORY PRACTICAL MANUAL WITH CONNECTION DIAGRAMS ★

Basic Electrical Engineering (BEE) & Measurements Lab Manual

Course Code:	R23-EEE-LAB1	**Academic Scheme:**	I B.Tech I Semester (R23 Scheme)
Verification:	Verified Authentic Academic Copy	**System Ref:**	STUDY-BLUEPRINT-VERIFIED-2026

DETAILED CURRICULAR MODULES & TOPICS

Experiment 1: Verification of KCL & KVL on Breadboard

Measuring branch currents and loop voltages with digital multimeters across resistive meshes.

Experiment 2: Verification of Thevenin's & Norton's Theorems

Determining open circuit voltage Vth, short circuit current Isc, and equivalent resistance Rth.

Experiment 3: Measurement of Active & Reactive Power in Single Phase Circuit

Using electrodynamometer wattmeter, ammeter and voltmeter to calculate power factor of inductive load.

Experiment 4: Swinburne's Test on DC Shunt Machine

Pre-determining efficiency of DC machine running as generator and motor at different loads.

Experiment 5: Calibration of Single-Phase Energy Meter

Determining percentage error of induction type energy meter using phantom loading method.

EVALUATION SCHEME & EXAMINATION BLUEPRINT

Continuous Lab Record	15 Marks
Viva-voce & Execution	15 Marks
University External Practical Exam	50 Marks (Total: 80 Marks)

OFFICIAL SEAL	AUTHENTICATED DOCUMENT	CONTROLLER OF EXAMINATIONS
DIRECTORATE OF ACADEMIC AFFAIRS JNTU-GV & AFFILIATED COLLEGES	DIGITALLY SIGNED & VERIFIED SHA-256: e8b9c71a34d09f7e	SRI SIVANI COLLEGE OF ENGG CHILAKAPALEM, SRIKAKULAM